Openings for the mass gap, and why it is not special to SU(3)

Recorded 2026-09-23 at the user's request and **parked**: the Planck-gap paper comes first, and none of these is being attacked now, except possibly the fourth, which is the link from the Planck-gap work. The note answers two questions the user asked and corrects one statement of the map.

Correction, 2026-10-02 (strong-side comparison). Opening 1's comparison now follows the maintained Wilson exponential-activity bound and the withdrawn Kogut–Susskind numerical threshold. Its own curvature normalization is still unchecked. This changes the cited strong-side baseline, not the status of the proposed curvature route.

The gap is expected for every compact simple non-abelian group. Lattice evidence shows gapped glueball spectra for $SU(N)$ from $N = 2$ to large N, with masses smooth in $1/N^2$, and for the other classical groups; the dividing line is between abelian and non-abelian, since compact $U(1)$ in four dimensions has a proven massless phase at weak coupling. $SU(3)$ fixes the constants; a method that works will almost surely be group-general.

A precision on the map. The position note says that the conjecture is T2′ together with T3, given T4, and the obligations map says that T3 presupposes T2′. T3 needs less: a gap $\delta_\infty(g) > 0$ on some interval $(0, g_1)$ along the scaling curve. T2′, a gap at every lattice coupling, is a property of a particular lattice action, and numerical Wilson-action evidence for a first-order bulk transition at $SU(N \geq 5)$ does not by itself identify the gapped set of the distinct Kogut–Susskind operator. For $SU(3)$ with the Wilson action the bulk behaviour is a crossover, so T2′ remains plausible there; the conditional theorem of the conditional-theorem note, stated along the trajectory, needs no T2′ at all.

Four openings, ranked by how directly they reach the place where the map ends, the control of about six blocking steps at order-one coupling.

1. Curvature: the Bakry–Émery criterion and its multiscale form.
2. The centre-stabilized theory on $\mathbb{R}^3 \times S^1$ at small radius.
3. Three dimensions first.
4. The Planck-gap link: Simon's valley lifting, a floor that holds for every state.

1. Evidence across groups

- $SU(N)$**.** Glueball masses and string tensions computed for N from 2 to about 12 are finite and extrapolate smoothly in $1/N^2$ (Lucini and Panero, Phys. Rep. **526**, 93, 2013, metadata; Lucini and Teper, JHEP **2001**(06), 050, metadata).
- G_2**.** The exceptional group has a trivial centre and still shows confinement behaviour and a first-order deconfinement transition (Pepe and Wiese, Nucl. Phys. B **768**, 21, 2007, metadata), so the centre is not what the gap rests on.
- $U(1)$**.** Four-dimensional compact $U(1)$ has a massless Coulomb phase at weak coupling (Guth, PRD **21**, 2291, 1980; Fröhlich and Spencer, CMP **83**, 411, 1982; metadata), which is why the abelian note separates the groups by the running, $b_0 = 0$ against $b_0 = 11N/(48\pi^2)$. In three dimensions compact $U(1)$ is gapped at every coupling (Polyakov, Nucl. Phys. B **120**, 429, 1977; proved by Göpfert and Mack, CMP **82**, 545, 1982; metadata), with a gap that vanishes in physical units in the continuum.

Bulk transitions depend on the action. With the standard Wilson plaquette action, "the bulk transition is a rapid cross-over if $N \leq 4$ but becomes an increasingly strong first order transition for $N \geq 5$"; for $N = 3$ a negative adjoint term avoids the bulk phase altogether (Rindlisbacher, Rummukainen and Salami, arXiv:2306.14319, passage). In the plane of fundamental and adjoint couplings, a line of first-order transitions ends at a critical point near the Wilson axis (Bhanot and Creutz, PRD **24**, 3212, 1981, metadata). A first-order transition can

involve finite-volume vacuum tunnelling splittings, but the cited Wilson-action evidence alone does not prove $\delta_\infty(g_b) = 0$ for the Kogut–Susskind operator; see the corrected coupling-set analysis §4b. Such intermediate-coupling behaviour does not by itself rule out a continuum gap along $g \to 0$. The statement the conjecture needs is $\mathcal{G} \supseteq (0, g_1)$ for some $g_1 > 0$, joined to T3 along the scaling curve.

Why $N \geq 5$ differs, and what sits beside the $SU(3)$ Wilson axis

Give the lattice action a fundamental coupling β_F and an adjoint coupling β_A. In the (β_F, β_A) plane a line of first-order bulk transitions ends at a critical point (Bhanot and Creutz 1981). For small N the Wilson axis $\beta_A = 0$ passes beside the endpoint and shows a rapid crossover; from $N = 5$ it crosses the line, and the transition strengthens with N. The large-N limit makes a non-analyticity between strong and weak lattice coupling unavoidable. In two dimensions, where the lattice theory is exactly solvable, it is analytic at every finite N and has a third-order transition at a critical 't Hooft coupling at $N = \infty$ (Gross and Witten, PRD **21**, 446, 1980; Wadia, PLB **93**, 403, 1980; metadata). In four dimensions the large-N transition is first order, and at finite N it survives as a genuine transition from $N = 5$ and is smoothed into a crossover below.

For $SU(3)$ the endpoint matters twice.

- **It realizes the second-order branch.** Approaching the endpoint, "the mass of the 0^{++} glueball decreases ... while the other glueball masses appear unchanged", which is "consistent with the notion that the bulk transition line ends in a critical endpoint with the continuum limit there being a ϕ^4 theory with a diverging correlation length only in the 0^{++} channel" (Heller, PLB **362**, 123, 1995; hep-lat/9508009, abstract). This is numerical evidence for a scalar critical-endpoint interpretation, not a constructed second continuum limit or a proof of uniqueness on families avoiding that point.
- **It sits beside the point where the map places H2.** The bands note scaled the continuum ratio $r_0 m_{0^{++}} = 4.21$ to $\beta_W = 5.7$ and found $\xi/a = 0.70$. Near there the Wilson axis passes beside the endpoint, where the scalar glueball softens, so the lattice ξ/a at $\beta_W \simeq 5.7$ is likely larger, and the measured lattice value should replace the scaled one. The conditional theorem allows the remedy: blocking generates adjoint and higher-representation terms anyway, so the reference interaction Φ_0 of H2 can be taken at a small negative adjoint coupling, away from the endpoint. For $N = 3$ such an action avoids the bulk phase, as Rindlisbacher, Rummukainen and Salami report from earlier work (arXiv:2306.14319, passage).

What the problem statement itself points to

Jaffe and Witten's official description, held as a local PDF, names in its §6.6 four routes to the gap (passage), and two of them are openings 1 and 4 below:

- a "duality transformation", with Debye screening in the Coulomb gas, proved through the sine-Gordon representation, as the model;
- the quartic potential $(A \wedge A)^2$ (Feynman, Nucl. Phys. B **188**, 479, 1981), which "may be tied to curvature in the space of connections" (Singer, Phys. Scripta **24**, 817, 1981): opening 1;
- "certain quantum mechanics problems with potentials involving flat directions ... do lead to bound states" (Simon 1983): opening 4;
- the $1/N$ expansion ('t Hooft 1974): gauge theory with $SU(N)$, $SO(N)$ or $Sp(N)$ "may be equivalent to a string theory with $1/N$ as the string coupling constant", which "might give a clear-cut explanation of the mass gap and confinement and perhaps a good starting point for a rigorous proof (for sufficiently large N)". Maldacena's paper is cited there as

"surprising progress along these lines for certain strongly coupled four-dimensional gauge systems with matter, but as of yet there is no effective approach to the gauge theory without fermions." Earlier, §6.1 allows that "there might some day be an asymptotic solution in a large N limit."

Chronology. Maldacena's conjecture was posted in November 1997 (hep-th/9711200); Witten's holographic account of non-supersymmetric Yang–Mills, in which "the spontaneous breaking of the center of the gauge group, magnetic confinement, and the mass gap are coded in classical geometry", in March 1998 (hep-th/9803131, abstract); supergravity glueball masses in agreement with lattice ratios in June 1998 (Csáki, Ooguri, Oz and Terning, hep-th/9806021, abstract); the Clay problems in May 2000. The statement was written with holography in hand, by one of its authors, and records no effective approach to the pure gauge theory. The usual reasons, not checked here against a primary text, are that the tractable supergravity regime is the opposite limit from the continuum one, with Kaluza–Klein modes at the glueball scale, and that it holds at large N only.

The large-N thread. It runs through Witten's own work: the $1/N$ expansion for baryons (Nucl. Phys. B **160**, 57, 1979) and for the $U(1)$ problem (Nucl. Phys. B **156**, 269, 1979), and the large-N lattice transition with Gross (1980), the non-analyticity behind the $N \geq 5$ bulk transition above. This programme records the route and does not adopt it: the conjecture is posed for every compact simple group, so a proof for sufficiently large N would leave $SU(3)$ open unless it came with explicit control down to $N = 3$; and at large N a lattice route meets the bulk transition, which it would have to avoid with a bulk-preventing action or with the centre-stabilized volume independence of opening 2, itself a large-N construction.

The panorama with $\hbar$ explicit. This repository writes the action as $S = -\frac{\hbar}{4g^2}\int F^a_{\mu\nu}F^{a\mu\nu}\,d^3x\,c\,dt$ with g dimensionless (G07, Proposition 7), so the weight $e^{iS/\hbar}$ contains g alone and $\hbar$ is hidden inside it: with a classical coupling $g_{\rm cl}$, $S = -\frac{1}{4g_{\rm cl}^2}\int\cdots$, one has $g^2 = \hbar g_{\rm cl}^2$. Three consequences follow.

- *The 't Hooft limit is a classical limit.* A diagram with E propagators, V vertices and genus h carries $(\hbar\lambda_{\rm cl})^{E-V}N^{2-2h}$ with $\lambda_{\rm cl} = g_{\rm cl}^2N$ ('t Hooft, Nucl. Phys. B **72**, 461, 1974, metadata). The loop parameter is $\lambda = \hbar g_{\rm cl}^2 N$, so $N \to \infty$ at fixed λ is $\hbar \to 0$ with $\hbar N$ fixed at fixed $g_{\rm cl}$, and $1/N^2 = (\hbar g_{\rm cl}^2/\lambda)^2$: the genus expansion is an expansion in $\hbar^2$. Yaffe formulates large-N limits in general as classical mechanics (Rev. Mod. Phys. **54**, 407, 1982, metadata). In holography it is literal: the bulk Newton constant in anti-de Sitter units is of order $1/N^2$, so the bulk loop expansion is the $1/N^2$ expansion.
- *The gap is non-perturbative in $\hbar$.* With the two-loop scale of the obligations map §5 and $b_0 = 11N/(48\pi^2)$, $mc^2 \propto \frac{\hbar c}{a}\exp(-\frac{24\pi^2}{11\,\hbar g_{\rm cl}^2 N})$ times a power of $\hbar g_{\rm cl}^2 N$: at fixed N, a and $g_{\rm cl}$ it vanishes faster than any power of $\hbar$, the way a tunnelling splitting does, which gives one precise reading of Jaffe and Witten's remark that the gap is "not classically visible". It depends on $\hbar$ and N only through $\hbar N$, so the 't Hooft limit is the classical limit along which the gap survives.
- *The small volume is a power of $\hbar$.* G07's zero-mode gap $\delta_1 g^{2/3}\hbar c/L$ is $\delta_1\hbar^{4/3}g_{\rm cl}^{2/3}c/L$: a floor of the Planck-gap kind, a power of $\hbar$ from the zero-point energy that lifts the valleys. The two regimes meet at the crossover $z \simeq 2$ of the valley note. Opening 4 lives in the power-law regime, where the Planck-gap methods apply; the infinite-volume gap lives in the non-perturbative one.

2. The openings

Opening 1: curvature

The configuration space of lattice gauge theory is a product of copies of the group, and a compact simple group carries positive Ricci curvature: for $SU(N)$ with the metric $\langle X, Y\rangle = \mathrm{Tr}(XY^*)$,

$\mathrm{Ric} = \frac{N}{2}\, g$. The Bakry–Émery criterion turns a positive lower bound on $\mathrm{Ric} + \mathrm{Hess}\, S$ into a log-Sobolev and a Poincaré inequality, that is into a spectral gap of the Langevin dynamics and exponential decay of correlations (Bakry and Émery, LNM 1123, 177, 1985, metadata). A flat group, $U(1)$, gets nothing from the first term, which lines up with the abelian failure.

Shen, Zhu and Zhu use exactly this to prove, for lattice Yang–Mills with 't Hooft coupling βN in any dimension $d > 1$, uniqueness of the infinite-volume measure, log-Sobolev and Poincaré inequalities and exponential decay of correlations, "a strictly positive mass gap", for $|\beta| < 1/(16(d-1))$ when the group is $SU(N)$ (CMP **400**, 2022; arXiv:2204.12737, abstract). In the Wilson normalization of this repository, $\beta_W = 2N/g^2$ with weight $e^{(\beta_W/N)\sum \mathrm{Re}\,\mathrm{Tr}\, U_p}$, their β is β_W/N^2, so for $SU(3)$ in four dimensions the condition is $\beta_W < 3/16$, that is

$$g^2 > 32.$$

That conversion is ours and must be checked against their definitions. If it stands, the curvature method already beats this repository's strong-coupling sufficient Wilson thresholds, $g^2 \geq 1059$ for the all-representation polymer bound (note) and $g^2 > 444$ by Dobrushin (note). The former Wilson figure 176 was a leading-activity estimate; the former Kogut–Susskind figure 388 was withdrawn on October 2 (note). The latter keeps an existential volume-uniform strong-coupling theorem and an explicit finite-volume bound, not a certified numerical uniform threshold. If the curvature conversion and physical transfer stand, the comparison narrows band B of the bands note from $\beta_W \in (0.0135, 5.7)$ to $\beta_W \in (0.19, 5.7)$.

The weak side is where the multiscale form enters. Bauerschmidt and Bodineau derive "a multiscale generalisation of the Bakry–Émery criterion" through the Polchinski renormalization equation, "effective for measures which are far from log-concave", and prove optimal log-Sobolev inequalities for the continuum sine-Gordon model at $\beta < 6\pi$ (CPAM **74**, 2020; arXiv:1907.12308, abstract). Its output is a spectral gap controlled at every scale, which is the form of hypothesis H1 in the conditional theorem. It also sits outside the family qualified in flow-conjugation: that note shows invariance when the same decomposition is conjugated; a new splitting or a curvature bound needs a separate analysis.

The risk is concrete. After a blocking step the effective action is a function on the coarse configuration manifold, and the criterion needs $\mathrm{Ric} + \mathrm{Hess}\, S_{\mathrm{eff}}$ bounded below at every scale. At weak coupling the Hessian is degenerate along gauge orbits and along the abelian valleys, and the curvature of the group has to carry those directions. **First note if opened:** check the conversion to $g^2 > 32$; compute the curvature budget for $SU(2)$ and $SU(3)$ against the Hessian of the Wilson action; and state exactly what lower bound one blocking step must preserve.

Opening 2: the centre-stabilized theory at small radius

Ünsal and Yaffe add a double-trace deformation that keeps centre symmetry unbroken; on $\mathbb{R}^3 \times S^1$ at small radius "the deformed Yang–Mills theory has a mass gap and exhibits linear confinement", by an analytic semiclassical treatment, and "there are no order parameters which distinguish the small and large radius regimes" (PRD **78**, 065035, 2008, abstract). The mechanism at small radius is Polyakov's monopole plasma, which Göpfert and Mack made rigorous for three-dimensional compact $U(1)$ at every coupling. **Target if opened:** a rigorous mass gap for the deformed theory at small radius, with explicit constants. That would be a theorem about a deformed theory; carrying it to $\mathbb{R}^4$ is the adiabatic-continuity conjecture.

Opening 3: three dimensions first

In three dimensions g^2 has the dimensions of energy, the theory is superrenormalizable, and the coupling at lattice scale is $g^2 a/\hbar c$, which vanishes as a power of a. The weak-side deficit of twelve orders of magnitude found in the small-field threshold note comes from the logarithmic running of four dimensions, and in three dimensions the ultraviolet steps shrink geometrically, so the

constant-chasing that failed in four dimensions may close. The one-function note reduces the problem to a lower bound $f(x) \geq f_- > 0$ on one function of one variable. A three-dimensional gap uniform in the spacing is open and would be a major result by itself.

Opening 4: the Planck-gap link

The small-volume mechanism of G07 and G08 is Simon's: along the flat valleys of the commutator potential the transverse modes are oscillators whose frequency grows with the distance along the valley, and their zero-point energy lifts the valley (Simon, Ann. Phys. **146**, 209, 1983, metadata). In its simplest form, for fixed x,

$$\frac{\hat{p}_y^2}{2m} + \frac{1}{2} m \Omega(x)^2 \hat{y}^2 \;\geq\; \frac{1}{2} \hbar\, \Omega(x),$$

an operator inequality that holds in every state. The torus-valley note finds the field-theory version term by term: the linear term $2|a|$ of the valley potential is the zero-point energy of the $k = 0$ charged modes.

The Planck-gap work sharpens what kind of floor this is. Its floors that rest on second moments of a prepared state fall to squeezing and to grid states, and the ones that survive rest on the commutator alone: the phase between Newton's polygon and the parabola, and the cost of disturbance. Simon's inequality is of the surviving kind, a spectral statement, so the valley lifting is the mass-gap instance of the lesson the Planck paper draws. **First step if opened:** hypothesis (H3) of the Feshbach note, the gap of the zero-mode Hamiltonian h_3 plus the even valley potential $U(a)$, proved by a Simon-type operator inequality with every constant explicit. That is the lower side of the small-volume corner S at weak coupling, and it stays within reach of the Planck-gap methods.

3. Consequence for STATE

The September 23 record left the mass-gap track paused; the subsequent September 26 user direction in STATE restored it as a main goal. This note records the four openings and the answer on groups: the gap is expected for every compact simple non-abelian group, and $SU(3)$ is the group that fixes the constants. The map's statement that the conjecture is T2′ with T3 is too strong: T3, with a gap on an interval $(0, g_1)$ along the scaling curve, is what is needed, and numerical Wilson-action bulk evidence does not establish failure of the Kogut–Susskind T2′ property. If one opening is taken up next, it is the fourth, which uses the Planck-gap methods directly; the first carries the most promise, and its first check is the conversion of Shen, Zhu and Zhu's threshold to $g^2 > 32$. The October 2 comparison uses the current Wilson sufficient bounds and records the withdrawn KS explicit threshold. It establishes no new curvature normalization, numerical uniform KS threshold or continuum gap.